

NAKAST — User Manual

Version 1.0 · Complete reference for running and interpreting NAKAST.

For a five-minute first run, see the [README](#). This manual assumes you have already installed the pipeline and covers what the README does not: how to read your results, what the pipeline assumes about your data, and where its limits are.

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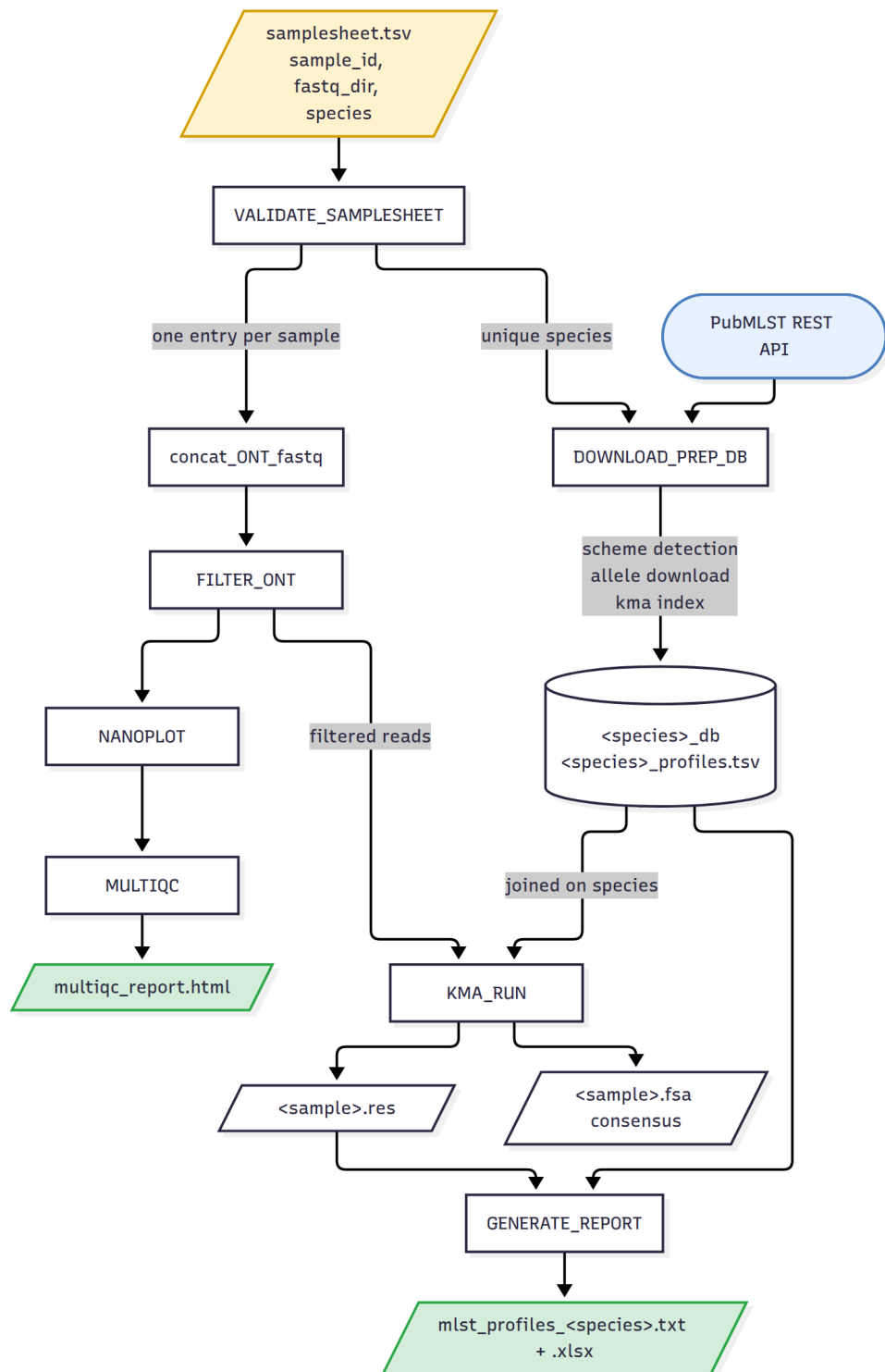
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1. What NAKAST does

NAKAST determines MLST profiles from Oxford Nanopore amplicon reads. You supply FASTQ directories and a species name; it returns a table of Sequence Types.



The species acts as a **grouping key**: the database is downloaded once per distinct species, each sample is matched to the database of its own species, and one report is produced per species. A single run may therefore mix species.

2. Background concepts

MLST (Multi Locus Sequence Typing) characterises a bacterial isolate by sequencing a small set of housekeeping gene fragments, usually seven. Each distinct sequence at a locus receives an integer **allele number**, curated centrally.

Allelic profile is the ordered set of allele numbers, for example 9-1-4-1-3-3-2.

Sequence Type (ST) is an integer assigned to each distinct allelic profile. A profile matches an ST only if *every* allele is an exact match to a known allele. One uncharacterised locus leaves the isolate untyped.

Clonal Complex (CC) groups related STs that share most of their alleles. Not every scheme defines clonal complexes: 102 of the 135 supported schemes do. The rest report CC = -, which is correct rather than a failure.

PubMLST (<https://pubmlst.org>) hosts the reference allele sequences and profile tables. NAKAST queries its REST API directly, so results reflect the current state of the database.

KMA is the aligner. It maps reads against a k-mer indexed database of allele sequences and reports identity, coverage and depth per template. It suits redundant amplicon data because it resolves which of many near-identical alleles best explains the reads.

3. A complete worked example

A run of three *Streptococcus agalactiae* isolates, from raw data to interpretation.

Step 1 — check your input data

Your basecalled, demultiplexed reads, one directory per sample:

```
ls ~/run_2026_03/fastq_pass/  
# barcode01 barcode02 barcode03  
ls ~/run_2026_03/fastq_pass/barcode01/ | head -3  
# FBD86797_pass_barcode01_0.fastq.gz
```

Step 2 — find the species name

```
nextflow run nakast.nf --list_species | grep -i agalactiae
# sagalactiae      1      7  yes  Streptococcus agalactiae
```

Step 3 — write the samplesheet

Three tab-separated columns:

```
printf 'sample_id\tfastq_dir\tspecies\n' > samplesheet.tsv
printf 'SGB11\t%s/barcode01\tsagalactiae\n' ~/run_2026_03/fastq_pass >>
samplesheet.tsv
printf 'SGB12\t%s/barcode02\tsagalactiae\n' ~/run_2026_03/fastq_pass >>
samplesheet.tsv
printf 'SGB13\t%s/barcode03\tsagalactiae\n' ~/run_2026_03/fastq_pass >>
samplesheet.tsv
```

Use absolute paths. Relative paths resolve against the directory you launch from.

Step 4 — run

```
nextflow run /path/to/nakast/nakast.nf \
  --samplesheet samplesheet.tsv \
  --outdir results_run2026_03
```

Expect roughly 5–10 minutes for three samples on a laptop, most of it in KMA. The first run adds a few minutes to build the Conda environments; later runs reuse them.

Step 5 — read the report

Sample	ST	CC	adhP	pheS	atr	glnA	sdhA	glcK	tkf	Profile
SGB11	10	cc12	9	1	4	1	3	3	2	OK
SGB12	24	cc452	5	4	4	3	2	3	3	OK
SGB13	2	cc1	1	1	3	1	1	2	2	OK

Three isolates typed: ST10, ST24 and ST2. Section 7 explains what to do when a row does not look like these.

Step 6 — check the QC

Open `results_run2026_03/qc/multiqc_report.html` to confirm read counts and quality distributions are as expected for the run.

4. Input specification

A tab-separated file with a header row and three columns.

Column	Required	Description
sample_id	yes	Unique identifier. Becomes the output file prefix
fastq_dir	yes	Directory containing .fastq.gz or .fastq files
species	yes	Short name from the catalogue, for example sagalactiae

Additional columns are ignored, so samplesheets written for earlier versions keep working.

Validation runs before any heavy work and checks that the three columns exist, that sample_id is non-empty and unique, that the referenced directories exist on disk, and that species appears in assets/pubmlst_species.tsv. A misspelled species fails in seconds with a suggestion rather than after a partial download.

A mixed-species batch is written exactly the same way:

```
sample_id  fastq_dir  species
SGB11     /data/run1/barcode01  sagalactiae
SAL07     /data/run1/barcode02  salmonella
```

5. Parameter reference

Input and output

Parameter	Default	Description
--samplesheet	samplesheet.tsv	Input table
--outdir	results	Base output directory
--qc_outdir	<outdir>/qc	QC outputs
--consensus_outdir	<outdir>/consensus_sequences	Consensus FASTA
--report_outdir	<outdir>/reports	.res files and profile tables
--mlst_report	mlst_profiles	Base name of the final report

Read filtering

Parameter	Default	Description
--nano_base_quality	20	Minimum Phred quality, passed to chopper
--min_length	300	

Parameter	Default	Description
		Minimum read length (filtlong)
--keep_percent	80	Keep this percentage of the best reads by bases (filtlong)

Allele calling

Parameter	Default	Description
--min_depth	30	Minimum depth, applied in KMA and again in the report script
--min_identity	90	Minimum identity to the reference allele (%)
--allele_coverage	90	Minimum allele (template) coverage (%)
--kma_k	31	K-mer size for the KMA index
--generate_consensus	true	Publish per-locus consensus sequences

Run modes

Parameter	Default	Description
--qc_only	false	Run QC only; no download, no typing
--list_species	false	Print the supported-species catalogue and exit
--help	false	Print usage and exit

Nextflow options worth knowing

These belong to Nextflow itself and take a single dash.

Option	Purpose
-resume	Reuse completed work from the previous run
-profile host	Use the tools on your PATH instead of Conda
-with-report	Extra HTML execution report

Resources

Assigned by label in `nextflow.config`. Edit that file to fit your machine.

Label	CPUs	Memory	Time	Processes
process_low	2	4 GB	2 h	validation, database, species listing
process_medium	6	10 GB	8 h	concatenation, filtering, QC, report
process_high	12	16 GB	24 h	KMA alignment

6. Output reference

```

results/
├─ mlst_profiles_<species>.txt      ← the answer
├─ mlst_profiles_<species>.xlsx
├─ qc/
│   └─ qc_<sample>/
│       └─ multiqc_report.html      ← run quality
├─ consensus_sequences/
│   └─ <sample>.fsa
├─ reports/
│   └─ <sample>.res                ← evidence per allele
│       └─ <species>_profiles.tsv
└─ pipeline_info/

```

Which file answers which question

Question	File
What ST is my isolate?	mlst_profiles_<species>.txt
Why was this allele called?	reports/<sample>.res
Did the run sequence well?	qc/multiqc_report.html
What sequence do I submit to PubMLST or confirm by Sanger?	consensus_sequences/<sample>.fsa
Which PubMLST version did this run use?	reports/<species>_profiles.tsv
How long did it take, what failed?	pipeline_info/

Final report columns

Column	Content
Sample	Sample identifier
ST	Sequence Type, or - if the profile matches nothing
CC	Clonal Complex, or - if unassigned or not defined by the scheme
one column per locus	Allele call with its quality annotation
Profile	OK, or INCOMPLETE (n/N) when fewer than N loci were recovered

Intermediate files

- `reports/<sample>.res` — raw KMA output. The authoritative record of identity, coverage, depth and score for every template considered.
- `reports/<species>_profiles.tsv` — the exact PubMLST profile table used. Keep it: PubMLST changes over time, and this file makes the run reproducible after the fact.
- `consensus_sequences/<sample>.fsa` — KMA consensus per locus.
- `pipeline_info/` — execution report, timeline and trace.

7. Interpreting your results

Work through this in order when a row is not a clean OK with an ST.

The allele annotations

Notation	Meaning
9	Exact match to allele 9
~9	Closest to allele 9, but the sequence differs
9?	Matches allele 9, but the read is longer than the reference
INS	The read covers less than the full allele
-	No call: below identity, coverage or depth thresholds

Only bare integers can match a PubMLST profile. This is the single most useful fact in this manual. If any locus carries ~, ?, INS or -, the profile cannot match and ST will be -. The ST lookup did not fail; the profile was simply incomplete.

Decision guide

ST = - but every locus has a bare integer. The profile is complete but that combination is not in PubMLST. This is a genuinely new ST. Consider submitting it.

ST = - and one locus shows ~N. The most common case. That locus differs from every known allele. Either it is a novel allele, or the consensus carries a sequencing error. To tell them apart, check the depth of that locus in the .res file. A consistent difference at high depth

points to a real novel allele; confirm by Sanger using the consensus in `consensus_sequences/`. A difference at low depth is more likely an artefact.

ST = - and one locus shows -. That locus was not recovered. Check its depth in the `.res` file. Depth below `--min_depth` means the amplicon underperformed, which is a wet-lab problem, not an analysis one. Re-amplify or sequence deeper.

Profile = INCOMPLETE (6/7). Six of seven loci were recovered. Same diagnosis as above: find the missing locus and check its depth.

ST assigned but CC = -. Usually correct. Many STs have no clonal complex, and 33 of the 135 schemes define none at all. Check the CC column in the catalogue for your species.

Every sample has ST = -. Suspect the species value before suspecting the data. Typing against the wrong scheme produces empty results without any error.

Reading the evidence file

```
column -t results/reports/SGB20.res | head
```

The columns that matter are `#Template` (locus and allele), `Template_Identity`, `Template_Coverage`, `Query_Coverage` and `Depth`. An allele called `~9` will show `Template_Identity` below 100. A locus reported as `-` may be absent from the file entirely, which means nothing passed the thresholds.

8. Assessing data quality

NAKAST does not enforce a quality gate beyond `--min_depth`. Use these reference points from validation on the in-house *S. agalactiae* amplicon panel.

Metric	Typical	Concerning
Depth per locus	several hundred to a few thousand	below 100
Loci recovered	7 of 7	6 or fewer
Annotated alleles per sample	0	2 or more
Median read quality	around Q18	below Q12

Depth varies a lot between loci in an amplicon panel: a ten-fold spread between the best and worst amplicon is normal. What matters is that the weakest locus clears `--min_depth`.

Very high depth does not rescue a bad amplification. If one locus is consistently weak across every sample in a run, the cause is the primer or the reaction, not the analysis.

9. Workflow architecture

Process	Tool	Purpose
VALIDATE_SAMPLESHEET	python, pandas	Pre-flight check of the samplesheet
DOWNLOAD_PREP_DB	python, kma	Detect scheme, download alleles and profiles, build index
concat_ONT_fastq	coreutils, gzip	Merge a barcode directory into one FASTQ
FILTER_ONT	filtlong, chopper	Length and quality filtering
NANOPLLOT	NanoPlot	Per-sample read QC
MULTIQC	MultiQC	Aggregate QC report
KMA_RUN	kma	Align reads against the allele database
GENERATE_REPORT	python, pandas, openpyxl	Allele calling, ST/CC assignment, reports
LIST_SPECIES	python	Print the supported-species catalogue

Failure behaviour differs by stage, on purpose:

Process	Strategy	Rationale
VALIDATE_SAMPLESHEET	terminate	A bad samplesheet should stop everything immediately
DOWNLOAD_PREP_DB	retry ×3, then finish	Survives transient network failures; lets QC finish so <code>-resume</code> can continue
KMA_RUN, GENERATE_REPORT	terminate	A silent failure here would produce a run with no results and exit 0
concat_ONT_fastq, FILTER_ONT, NANOPLLOT, MULTIQC	ignore	One bad sample should not abort the batch

10. How allele calling works

Step 1 — depth filter

Templates below `--min_depth` are discarded, both by KMA and again in the report script.

Step 2 — pick a winner per locus

Several alleles of the same locus usually attract reads, since they differ by a few bases. NAKAST ranks them with a confidence score:

```
Score_Ratio = Score / Expected
Confidence  = Score_Ratio × log1p(Depth) × perfect_bonus
```

`perfect_bonus` is 1.5 when identity, template coverage and query coverage are all 100%, and 1.0 otherwise. The highest-confidence template wins the locus.

The bonus is multiplicative rather than absolute on purpose: a perfect match still needs depth behind it, so a single well-aligned read cannot outrank a deeply covered allele.

Step 3 — annotate the call

Notation	Condition
9	Identity, template coverage and query coverage all 100%
~9	Query and template coverage 100%, identity below 100%
9?	Query coverage above 100%, identity and coverage above thresholds
INS	Query coverage below 100%
-	Identity or template coverage below thresholds, or no template passed

A note on `?`: query coverage above 100% means the read is *longer* than the template. With an amplicon panel this reflects read-through rather than truncation, so the call is usually sound. Treat `9?` as “probably allele 9, worth confirming” rather than as a failure.

11. ST and CC assignment

The allele calls are joined into a signature (9/1/4/1/3/3/2) and matched against the same signature built from the PubMLST profile table. The match is exact and string-based:

- Loci come from `<species>_loci.txt`, written from the scheme definition, then ordered as they appear in the profile table so the report preserves PubMLST's canonical order.
- The profile table is read as text so a missing value cannot silently turn a locus column into floating-point numbers and produce `1.0/1/2/...`, which would fail every match.
- The CC column is detected tolerantly (`clonal_complex`, `clonal complex`, `cc`, `clonalcomplex`). When absent, `CC = -`.
- No match returns `ST = -`. **NAKAST never reports a nearest or approximate ST.**

12. Species support

`assets/pubmlst_species.tsv` and `docs/supported_species.md` catalogue the **135 PubMLST schemes** that define ST profiles. Regenerate them with:

```
python3 bin/list_pubmlst_species.py --write-catalogue .
```

Two assumptions that hold for *S. agalactiae* but not in general, which NAKAST therefore resolves at run time:

- **The MLST scheme is not always scheme 1.** Seven schemes use a different id; *Salmonella* uses 2. NAKAST selects the scheme whose description starts with MLST.
- **Schemes do not always have seven loci.** They range from 2 to 10, and 48 of 135 differ from seven. The locus list comes from the scheme definition, never from a fixed list.

Locus names are also less regular than they look: `MLST_adk`, `16S_rRNA` and `int_hyp` all contain underscores, so allele identifiers are split from the right.

13. Assumptions

These matter for interpreting results.

1. **Amplicon sequencing, not whole genome.** Reads are assumed to come from targeted amplification of the MLST loci. Depth is expected in the hundreds or thousands and the default thresholds reflect that. Whole-genome data will give far lower per-locus depth and many - calls.
2. **The species is known and declared.** NAKAST does not identify the organism; it types the sample against the scheme you specify. A wrong species produces an empty result, not an error.
3. **PubMLST is reachable and authoritative.** The database is downloaded fresh on every run, so results depend on the state of PubMLST at run time. That is why the profile table is published with the outputs.
4. **The scheme defines ST profiles.** Schemes without a primary key, such as cgMLST, are excluded from the catalogue and rejected at download time.
5. **Sample identifiers are unique** across the samplesheet. They name the output files.
6. **Reads are basecalled and demultiplexed.** NAKAST starts from per-barcode FASTQ. It does not basecall, demultiplex, or trim adapters and barcodes.
7. **One organism per sample.** Mixed or contaminated cultures produce competing alleles at some loci; the confidence score will pick one, and a ~ annotation is often the only visible hint.

14. Limitations

1. **Oxford Nanopore only.** There is no short-read path. Filtering and alignment parameters are tuned for the ONT error profile, and the samplesheet takes a directory of FASTQ files rather than read pairs.
2. **No novel allele detection or submission.** A sequence absent from PubMLST is reported as ~N against the closest known allele. NAKAST will not flag it as new, propose a number, or prepare a submission. Confirmation, by Sanger for instance, is a manual step; the consensus in consensus_sequences/ is the starting point.
3. **Network required.** There is no offline mode and no local cache between runs. Every run re-downloads the scheme.

4. **Biologically validated on one species.** The architecture is species-agnostic and was exercised against the *Salmonella*, *Brucella*, *Vibrio* and *Mycoplasma genitalium* schemes, but validation against known isolates was done only for *S. agalactiae* with the in-house amplicon panel. Alignment parameters are tuned for that panel.
5. **Clonal complexes depend on the scheme.** 33 of 135 schemes define none, and even where defined many STs have no CC.
6. **No contamination or mixed-sample detection.** There is no explicit check for multiple alleles at a locus.
7. **Depth thresholds are global.** `--min_depth` applies to every locus equally. A panel where one amplicon systematically underperforms may need the threshold lowered for the whole run.
8. **One report per species.** Batches spanning several species produce several files rather than one merged table, because the locus columns differ.

15. Troubleshooting

Unknown configuration profile: 'conda' — Conda is enabled by default; drop the `-profile conda` flag.

command not found / exit status 127 — the process ran against the host PATH instead of its Conda environment. Confirm `nextflow config .` reports `conda { enabled = true }`, and that no stray `nextflow.config` in your launch directory is overriding it.

Variable declarations cannot be mixed with config statements — a `nextflow.config` in your launch directory uses syntax that Nextflow 26 rejects. Nextflow merges the config from the launch directory with the pipeline's own, so an unrelated config from an old project will break the run. Launch from a clean directory.

Every ST is - — check the `species` value first, then the allele columns. See [section 7](#).

Species rejected by validation — run `--list-species` and use the exact short name. The error message suggests near matches.

Download fails — the pipeline retries three times before stopping, and lets in-flight QC finish. Re-run with `-resume` when connectivity returns; completed work is reused.

Nothing is cached on -resume — check that nothing in the task inputs changes between runs. Interpolating a Path object such as launchDir directly into a process script gives it an unstable hash; pass it as a val input instead.

A sample is missing from the report — its filtering or QC step failed and was ignored so the batch could continue. Look for Error is ignored in the Nextflow log and check pipeline_info/ for which process failed.

16. Reproducibility

Every process declares its own Conda environment with exact versions:

Process	Environment
VALIDATE_SAMPLESHEET	python=3.13.11 pandas=2.3.3
DOWNLOAD_PREP_DB, KMA_RUN	python=3.13.11 kma=1.6.8
concat_ONT_fastq	python=3.13.11 coreutils=9.5 gzip=1.12
FILTER_ONT	python=3.13.11 filtlong=0.3.1 chopper=0.12.0 gzip=1.12
NANOPLLOT	python=3.13.11 nanoplott=1.46.2
MULTIQC	python=3.13.11 multiqc=1.33
GENERATE_REPORT	python=3.13.11 pandas=2.3.3 numpy=2.4.0 openpyxl=3.1.5
LIST_SPECIES	python=3.13.11

Environments are cached in \$HOME/.nextflow/conda and built with mamba. Identical specifications share one environment.

The one input that is *not* pinned is PubMLST itself, which changes as curators add alleles and STs. To make a past run reproducible, keep reports/<species>_profiles.tsv and the pipeline_info/ trace alongside the report.

17. Design notes

Two findings from tuning against real *S. agalactiae* data are recorded here because they are counter-intuitive and easy to reintroduce. This section is aimed at maintainers rather than users.

KMA's -mrc is not allele coverage. -mrc is *minimum query coverage*: the fraction of the **read** that must align. An earlier version passed --allele_coverage to it. With reads averaging around 950 bp against 500 bp amplicons, most reads cannot reach 90% query coverage. Loci with abundant reads survived anyway, but weakly amplified loci lost every read and vanished despite being perfect matches at 100% identity and coverage. The flag was removed; allele coverage is enforced downstream on Template_Coverage, where it belongs.

filtlong's quality thresholds are not on the Phred scale. --min_mean_q and --min_window_q take a 0–100 accuracy percentage, so passing 20 filters nothing at all (Q20 corresponds to 99). Setting them to a value that does filter proved harmful: at 95 the weakly amplified loci lost all support, because low-abundance loci also tend to carry the lowest-quality reads. The flags were removed and quality is controlled solely by chopper, on the Phred scale, through --nano_base_quality.

A related negative result: escalating quality tiers (Q30 → Q25 → Q20) is not viable for typical ONT amplicon data. With a median per-read quality near Q18, thresholds above roughly Q22 retain a fraction of a percent of bases, and the surviving fragments are shorter than a single amplicon, so they cannot cover an allele.

18. Support, citation and licence

Reporting problems. Open an issue at <https://github.com/javier-tognarelli/nakast/issues>. Include the Nextflow version (nextflow -version), the command you ran, and the relevant part of .nextflow.log.

Authorship.

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Citing NAKAST. Cite this pipeline together with PubMLST and the underlying tools. The complete list of third-party components and their licences is in THIRD PARTY LICENSES.md.

Data source. Allele and profile definitions come from PubMLST (<https://pubmlst.org>), hosted at the University of Oxford. Acknowledge PubMLST and the relevant scheme in any publication using these results.